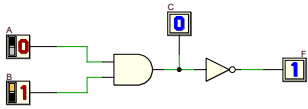


2 input AND gate

Logic expression  
 $F = AB$

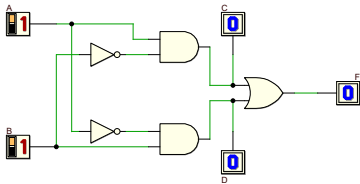
| Input |   | Output |
|-------|---|--------|
| A     | B | F      |
| 1     | 1 | 1      |
| 1     | 0 | 0      |
| 0     | 1 | 0      |
| 0     | 0 | 0      |



2 input NAND gate

Logic expression  
 $F = (AB)\text{bar}$

| Input |   | Output |   |
|-------|---|--------|---|
| A     | B | C      | F |
| 1     | 1 | 1      | 0 |
| 1     | 0 | 0      | 1 |
| 0     | 1 | 0      | 1 |
| 0     | 0 | 0      | 1 |



2 input XOR gate

Logic expression  
 $F = A\text{bar}B + AB\text{bar}$

| Input |   | Output |   |   |
|-------|---|--------|---|---|
| A     | B | C      | D | F |
| 1     | 1 | 0      | 0 | 0 |
| 1     | 0 | 1      | 0 | 1 |
| 0     | 1 | 0      | 1 | 1 |
| 0     | 0 | 0      | 0 | 0 |